Battery Monitor

Release 1.0

TekTutor

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This project documents a 12 V battery monitor.

CHAPTER

ONE

OVERVIEW

The battery monitor reads the pack voltage, converts it into a state of charge and warns the driver when the pack runs low.

1.1 Voltage to state of charge

Pack voltage	State of charge	Low warning
12700 mV	100 %	No
12250 mV	50 %	No
11999 mV	22 %	Yes
11800 mV	0 %	Yes

Sphinx is published at https://pypi.org/project/Sphinx/ and Sphinx-Needs at https://pypi.org/project/sphinx-needs/.

CHAPTER

TWO

REQUIREMENTS

Requirement: **Report state of charge** *REQ_SOC*

verified
ASIL: B

Issue: BMS-101

satisfied by: *SPEC_SOC_LINEAR*
verified by: *TC_SOC_CLAMP*, *TC_SOC_RANGE*

The system shall report the state of charge of the 12 V pack as a whole percentage from 0 to 100.

Requirement: **Warn on low battery** *REQ_LOW_WARN*

in_progress
ASIL: B

Issue: BMS-102

satisfied by: *SPEC_LOW_THRESHOLD*
verified by: *TC_LOW_BELOW*, *TC_LOW_EDGE*

The system shall warn the driver when the pack voltage falls below the low threshold.

Requirement: **Log voltage samples** *REQ_LOG*

untested
ASIL: QM

Issue: BMS-103

satisfied by: *SPEC_LOG_FORMAT*

The system shall store every voltage sample with a timestamp for workshop diagnostics.

CHAPTER

THREE

SPECIFICATIONS

Specification: **Linear voltage to SOC mapping** *SPEC_SOC_LINEAR*

accepted

satisfies: *REQ_SOC*

Map 11800 mV to 0 % and 12700 mV to 100 % on a straight line. Clamp every value outside that range to 0 % or 100 %.

Specification: **Low threshold of 12000 mV** *SPEC_LOW_THRESHOLD*

accepted

satisfies: *REQ_LOW_WARN*
mitigates: *RISK_DEEP_DISCHARGE*

Report the pack as low when the last sample is below 12000 mV. A sample of exactly 12000 mV is not low.

Specification: **Sample log format** *SPEC_LOG_FORMAT*

draft

satisfies: *REQ_LOG*

Store each sample as `<ISO 8601 timestamp>;<voltage in mV>`.

CHAPTER

FOUR

TEST CASES

Each test case below has a matching function in `tests/test_monitor.py`.

Test Case: **SOC at mid range** *TC_SOC_RANGE*

Result: pass

Source: tests/test_monitor.py

verifies: *REQ_SOC*

Feed 12250 mV. Expect 50 %.

Test Case: **SOC clamps outside range** *TC_SOC_CLAMP*

Result: pass

Source: tests/test_monitor.py

verifies: *REQ_SOC*

Feed 13500 mV and expect 100 %. Feed 10000 mV and expect 0 %.

Test Case: **Low warning below threshold** *TC_LOW_BELOW*

Result: pass

Source: tests/test_monitor.py

verifies: *REQ_LOW_WARN*

Feed 11999 mV. Expect the low warning.

Test Case: **No warning at threshold** *TC_LOW_EDGE*

Result: not_run

Source: tests/test_monitor.py

verifies: *REQ_LOW_WARN*

Feed exactly 12000 mV. Expect no low warning.

CHAPTER

FIVE

RISK REGISTER

Risk: **Deep discharge damages the pack** *RISK_DEEP_DISCHARGE*

open

Severity: high

mitigated by: *SPEC_LOW_THRESHOLD*

If the driver gets no warning, the pack can discharge below 10.5 V and lose capacity for good.

Risk: **Voltage sensor drift** *RISK_SENSOR_DRIFT*

open

Severity: medium

An aging sensor can read 100 mV high and delay the low warning.

5.1 Risks without mitigation

ID	Title	Severity	Status
*RISK_SENSOR_*	Voltage sensor drift	medium	open

CHAPTER

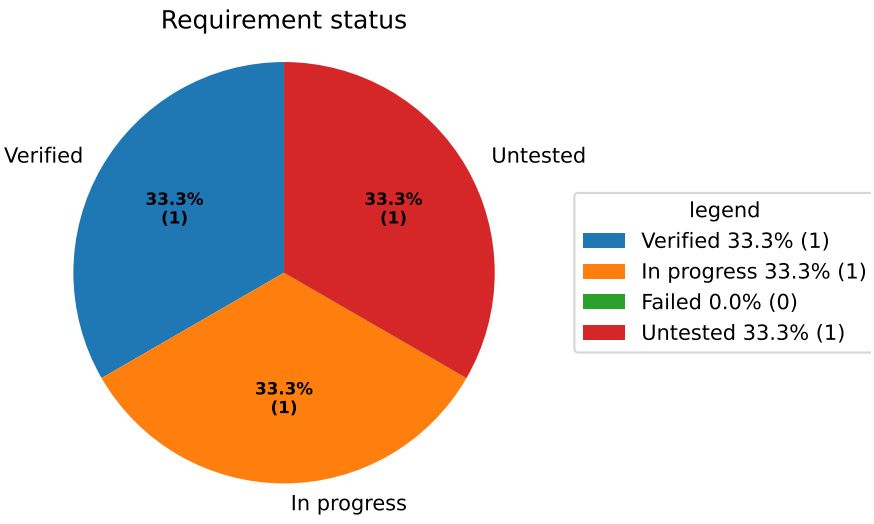
SIX

TRACEABILITY

6.1 Requirement to test matrix

ID	Title	Status	Specifications	Test cases	Issue
REQ_LOG	Log voltage samples	untested	*SPEC_LOG_*		BMS-103
*REQ_LOW_*	Warn on low battery	in_progress	*SPEC_LOW*	*TC_LOW_B* *TC_LOW_E*	BMS-102
REQ_SOC	Report state of charge	verified	*SPEC_SOC_*	*TC_SOC_C* *TC_SOC_R*	BMS-101

6.2 Requirement status



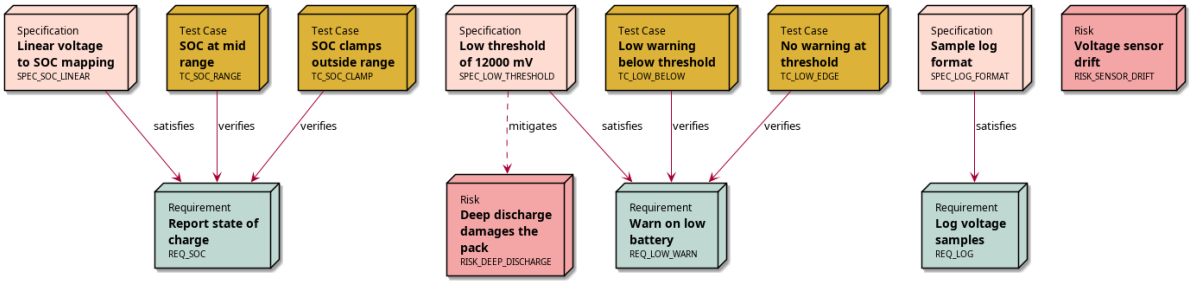


Fig. 1: Battery monitor traceability

6.3 Traceability graph

6.4 All needs

ID	Type	Title	Status	Tags
REQ_LOG	req	Log voltage samples	untested	diagnostics
REQ_LOW_W.	req	Warn on low battery	in_progress	warning; driver
REQ_SOC	req	Report state of charge	verified	soc; display
RISK_DEEP_I	risk	Deep discharge damages the pack	open	
RISK_SENSOI	risk	Voltage sensor drift	open	
SPEC_LOG_F	spec	Sample log format	draft	diagnostics
SPEC_LOW_T	spec	Low threshold of 12000 mV	accepted	warning
SPEC_SOC_L	spec	Linear voltage to SOC mapping	accepted	soc
TC_LOW_BEI	test	Low warning below threshold		
TC_LOW_EDC	test	No warning at threshold		
TC_SOC_CLA	test	SOC clamps outside range		
TC_SOC_RAN	test	SOC at mid range		

6.5 Open work

SPEC_LOG_FORMAT: Sample log format (draft) [diagnostics]
RISK_DEEP_DISCHARGE: Deep discharge damages the pack (open)
RISK_SENSOR_DRIFT: Voltage sensor drift (open)
REQ_LOW_WARN: Warn on low battery (in_progress) [warning; driver]
REQ_LOG: Log voltage samples (untested) [diagnostics]

CHAPTER

SEVEN

COVERAGE GAPS

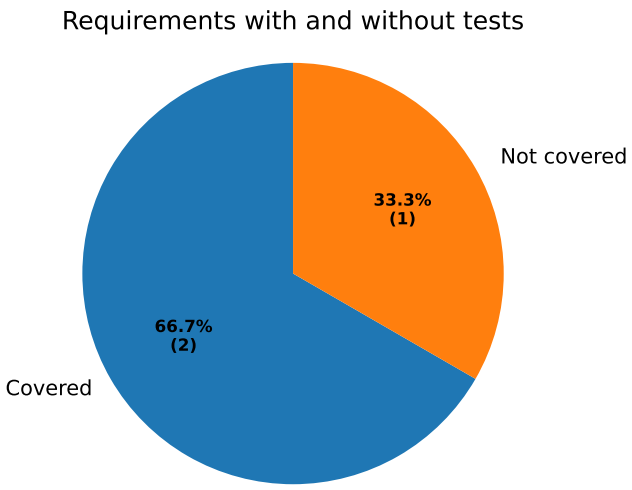
7.1 Requirements without any test case

ID	Title	Issue
REQ_LOG	Log voltage samples	BMS-103

7.2 Requirements without a specification

No needs passed the filters

7.3 Test coverage of requirements



7.4 Requirements whose tests have not all passed

ID	Title	Status	Test cases
REQ_LOW_WAR	Warn on low battery	in_progress	*TC_LOW_BELOW*; *TC_LOW_EDGE*